CS-E4740 - Federated Learning

L3 - FL Algorithms I

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Calendar **Glossary** **Book** **GitHub**

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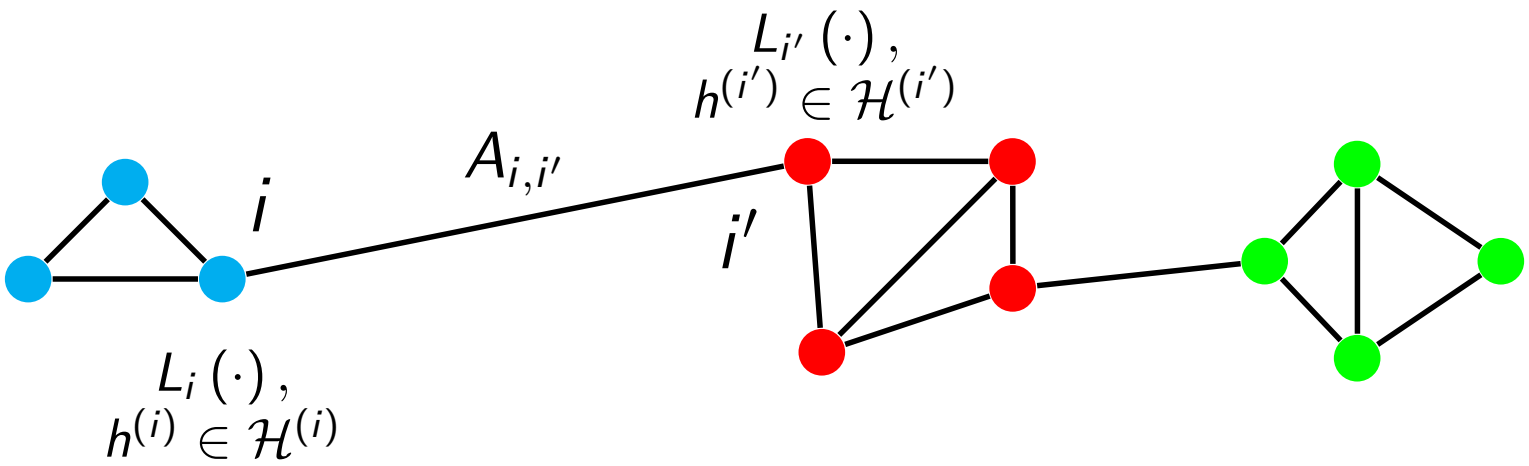
Recap: FL Network



- FL network with devices $i = 1, \ldots, n$ (graph with n nodes).
- Some connected by edge $\{i, i'\} \in \mathcal{E}$ with weight $A_{i,i'} > 0$.
- Device i learns a hypothesis $h^{(i)} \in \mathcal{H}^{(i)}$, e.g., linear model

$$h^{(i)}(\mathbf{x}) = \mathbf{x}^{\top}\mathbf{w}^{(i)}, \qquad \mathbf{x}, \mathbf{w}^{(i)} \in \mathbb{R}^{d}.$$

- Usefulness of $h^{(i)}$ measured by local loss function $L_i(\cdot)$.

Recap: Federated Learning Design Principle

- Balance the local loss with collaboration between devices via the generalized total variation minimization (GTVMin)

$$\min_{\substack{h^{(1)} \in \mathcal{H}^{(1)} \\ \vdots \\ h^{(n)} \in \mathcal{H}^{(n)}}} \sum_{i=1}^{n} L_i \left(h^{(i)}\right) + \alpha \sum_{\{i,i'\} \in \mathcal{E}} A_{i,i'} d^{(h^{(i)}, h^{(i')})}. \qquad \text{(GTVMin)}$$

- $\alpha > 0$ controls trade-off between local loss and enforcing similarity between connected nodes.
- d measures discrepancy measure between the local hypotheses of neighboring nodes

Recap: Fixed-Point Characterization

A necessary condition for a (GTVMin) solution $\left(\widehat{h}^{(1)}, \ldots, \widehat{h}^{(n)}\right)$:

$$\widehat{h}^{(i)} = \mathcal{F}^{(i)}\left(\widehat{h}^{(1)}, \ldots, \widehat{h}^{(n)}\right) \qquad \forall i = 1, \ldots, n, \qquad \text{(NecCond)}$$

where the operators $\mathcal{F}^{(i)} : \mathcal{H}^{(i)} \rightarrow \mathcal{H}^{(i)}$ are given by

$$\mathcal{F}^{(i)}\left(\widehat{h}^{(1)}, \ldots, \widehat{h}^{(n)}\right) := \underset{h \in \mathcal{H}^{(i)}}{\arg\min}\, L_i\left(h\right) + \alpha \sum_{j \in \mathcal{N}^{(i)}} A_{i,j} d^{(h, \widehat{h}^{(j)})}.$$

Note that

- $\mathcal{F}^{(i)}$ depends on local dataset via $L_i\left(\cdot\right)$,
- $\mathcal{F}^{(i)}$ depends on FL network via $\mathcal{N}^{(i)}$ and $A_{i,i'}$.

A Blueprint for FL Algorithms

- Assume that we found a fixed-point characterization of GTVMin with operators $\mathcal{F}^{(i)} : \mathcal{H}^{(i)} \rightarrow \mathcal{H}^{(i)}$, $i = 1, \ldots, n$.
- Induces iterative alg.: at time t, each device i computes

$$\widehat{h}^{(i,t+1)} = \mathcal{F}^{(i)}\big(\widehat{h}^{(1,t)}, \ldots, \widehat{h}^{(n,t)}\big) \qquad \text{(FLAlgo)}$$

- Update needs to done *synchronously* by all devices.
- Next lecture extends (FLAlgo) to allow *asynchrony*.

From ML to Optimization to Fixed-Point Operators

- ERM: find optimal model parameters $\mathbf{w}^\star$ min. a loss f:

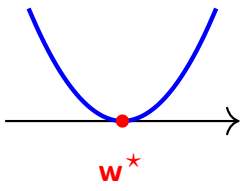
$$\min_{\mathbf{w}\in\mathbb{R}^d} f(\mathbf{w}) \quad \rightsquigarrow \quad \text{optimal solution } \mathbf{w}^\star \in \mathbb{R}^d.$$

- Fixed point approach: identify operator $\mathcal{F} : \mathbb{R}^d \to \mathbb{R}^d$ s.t.

$$\mathbf{w}^\star = \mathcal{F}(\mathbf{w}^\star).$$

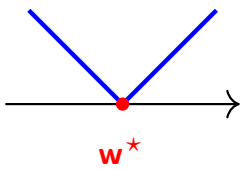
Case 1: smooth & convex f
(zero-gradient condition)

$$\nabla f(\mathbf{w}^\star) = \mathbf{0}$$
$$\iff \mathbf{w}^\star - \eta \nabla f(\mathbf{w}^\star) = \mathbf{w}^\star$$



Case 2: Continuous & convex f
(proximal methods)

$$\mathbf{w}^\star = \mathbf{prox}_{\eta f}(\mathbf{w}^\star)$$

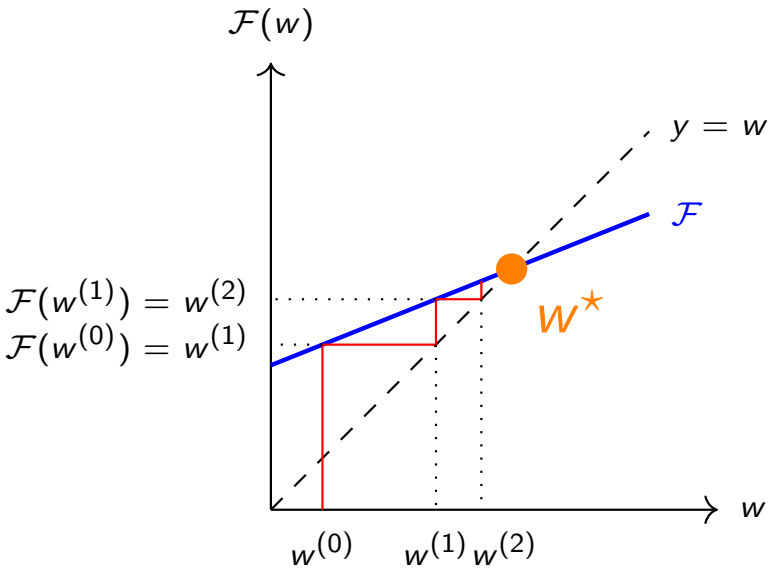


Fixed-Point Iterations

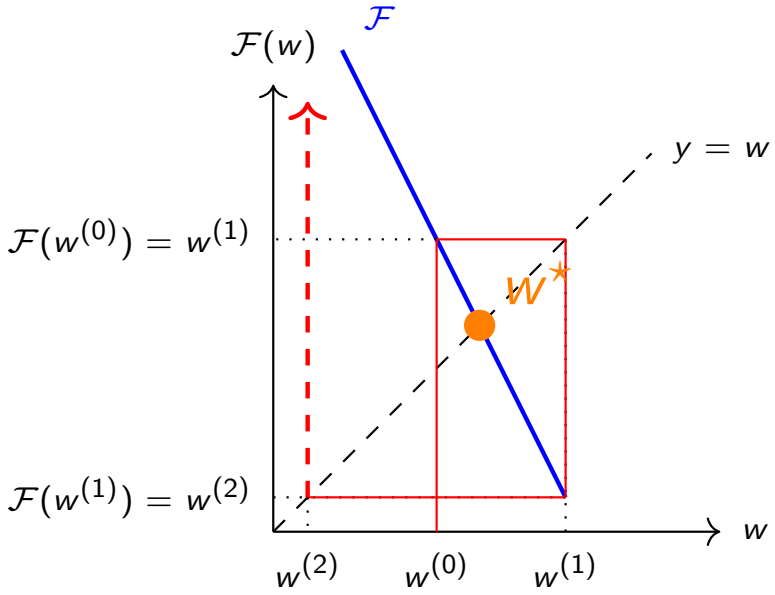
We can try to find $w^\star$ by applying the operator $\mathcal{F}$ repeatedly:

$$w^{(t+1)} = \mathcal{F}(w^{(t)})$$

Convergence



Divergence



Depending on properties of f and step size $\eta > 0$, we may obtain contractions or firmly non-expansive operator and convergence of fixed-point iterations $w^{(t+1)} = \mathcal{F}(w^{(t)})$ to $w^\star$.

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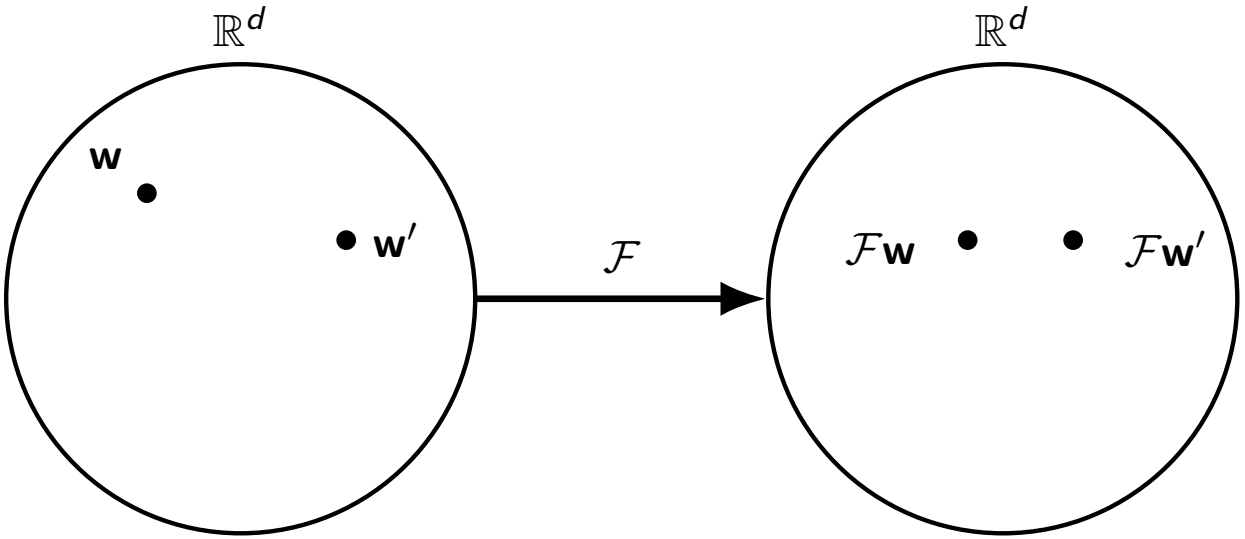
Contractions

A function $\mathcal{F} : \mathbb{R}^d \to \mathbb{R}^d$ is a contraction (or contractive) with respect to a norm $\|\cdot\|$, if there exists a $\kappa \in [0, 1)$, such that

$$\|\mathcal{F}(\mathbf{w}) - \mathcal{F}(\mathbf{w}')\| \leq \kappa \|\mathbf{w} - \mathbf{w}'\| \qquad \forall \mathbf{w}, \mathbf{w}' \in \mathbb{R}^d.$$



$\mathcal{F}$ shrinks distance between any two points by at least κ.

Contractions: Remarks

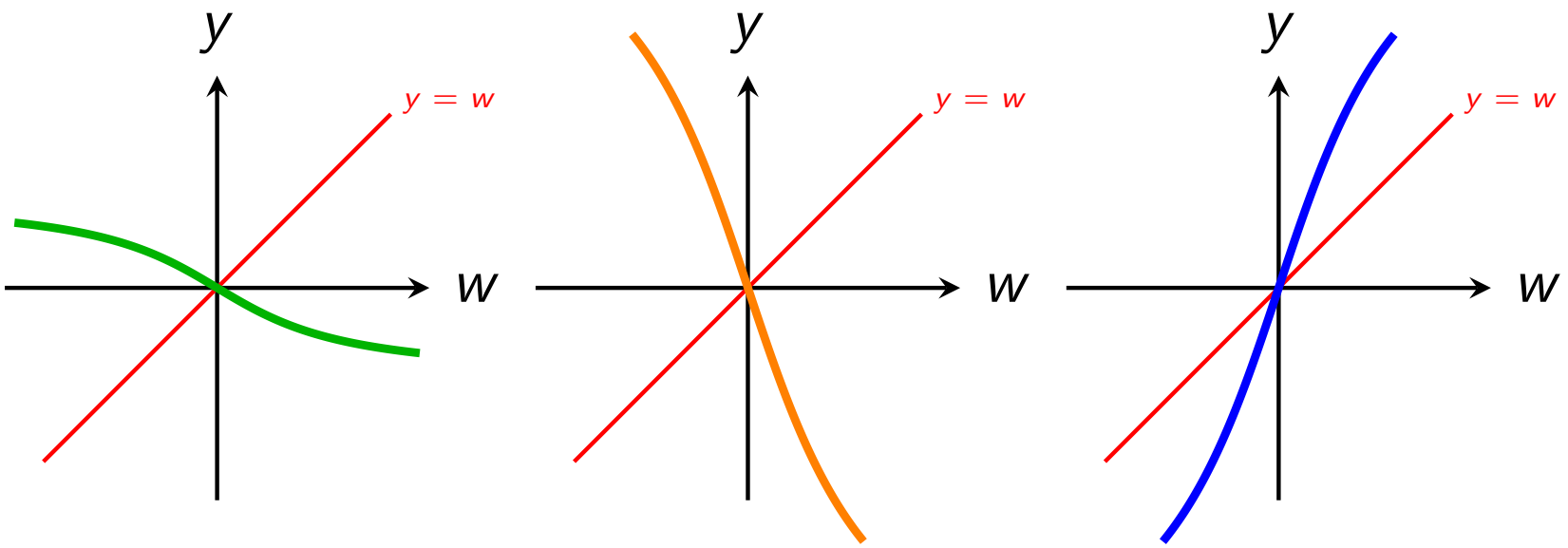
- The notion of a contractive operator generalizes naturally from $(\mathbb{R}^d, \|\cdot\|)$ to arbitrary metric spaces.
- We may also consider functions $\mathcal{F} : \mathcal{W} \to \mathbb{R}^d$, $\mathcal{W} \subset \mathbb{R}^d$.
- *Norm-dependency:* An operator $\mathcal{F}$ may be a contractive operator for one norm $\|\cdot\|$, but not w.r.t. another norm.
- Often: Euclidean norm induced by standard inner product

$$\|\mathbf{w}\|_2 := \sqrt{\sum_{j=1}^{d} w_j^2} = \sqrt{\langle \mathbf{w}, \mathbf{w} \rangle}$$

for a vector $\mathbf{w} = (w_1, \ldots, w_d)^\top \in \mathbb{R}^d$ in Euclidean space.

Contractions on $\mathbb{R}$: Intuition

$$|\mathcal{F}(w) - \mathcal{F}(w')| \leq \kappa |w - w'| \text{ with } \kappa \in [0, 1) \quad \forall w, w' \in \mathbb{R}$$



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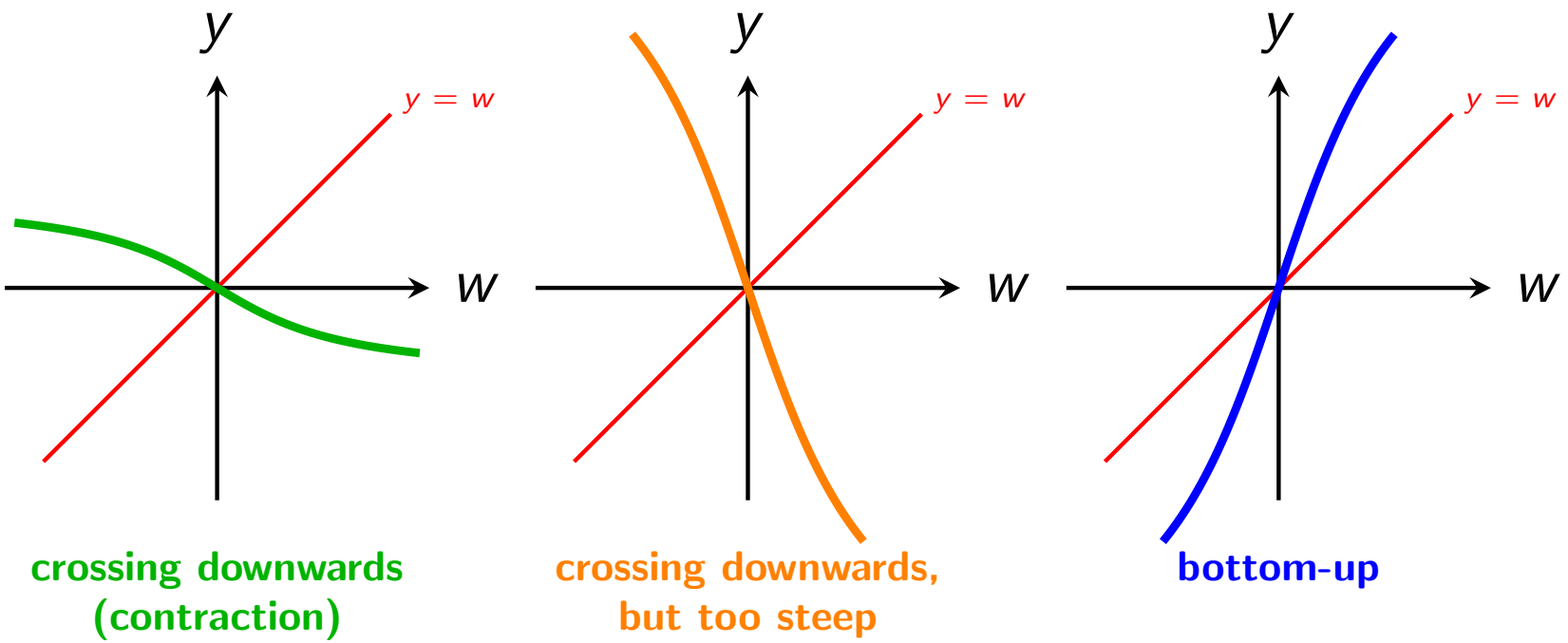


- Contractions can *only* cross the identity line $y = w$ downwards.
- Not every downward crossing is a contraction (depending on slope).

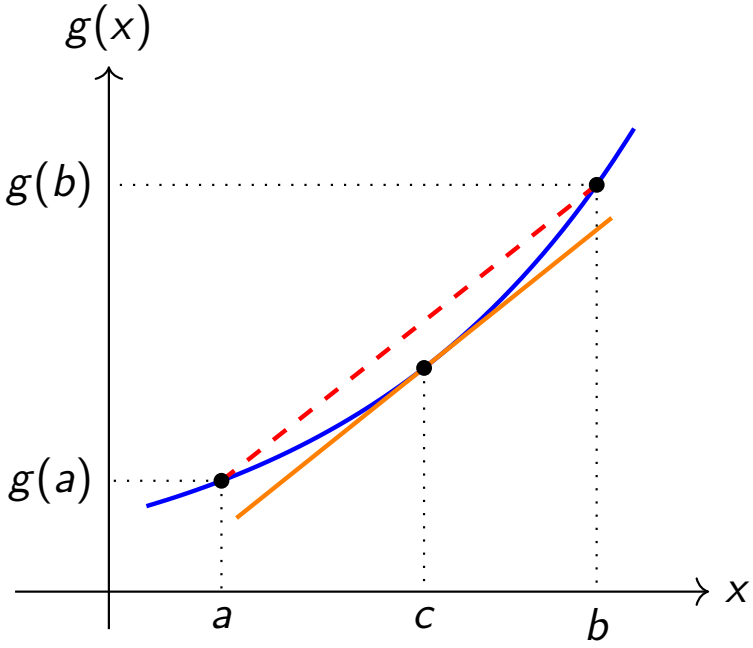
The Mean Value Theorem

Let $g : [a, b] \to \mathbb{R}$ be a continuous function on $[a, b] \subset \mathbb{R}$ that is differentiable on (a, b). Then, there exists $c \in (a, b)$ s.t.

$$g'(c) = \frac{g(b) - g(a)}{b - a}.$$

Geometric Interpretation

There exists $c \in (a, b)$ where the tangent to the graph is parallel to the secant line passing through $(a, g(a))$ and $(b, g(b))$.



Contractions on $\mathbb{R}$: Formal Characterization

Lemma

A differentiable function $\mathcal{F} : \mathbb{R} \to \mathbb{R}$ is a contractive operator w.r.t. $|\cdot|$ with constant κ if and only if $\sup_{w \in \mathbb{R}} |\mathcal{F}'(w)| \leq \kappa < 1$.

Proof.

($\Leftarrow$) By the *mean value theorem*: for any $w, w' \in \mathbb{R}$ (w.l.o.g. assume $w < w'$) there exists a $c \in (w, w')$

$$|\mathcal{F}(w) - \mathcal{F}(w')| = \underbrace{|\mathcal{F}'(c)|}_{\leq \kappa} |w - w'| \leq \kappa |w - w|.$$

($\Rightarrow$) As $|\mathcal{F}(w+h) - \mathcal{F}(w)| \leq \kappa |w + h - w| = \kappa |h|$, for $h \neq 0$:

$$|\mathcal{F}'(w)| = \lim_{h \to 0} \frac{|\mathcal{F}(w+h) - \mathcal{F}(w)|}{|h|} \leq \kappa,$$

Linear Contractions on $\mathbb{R}^d$

Lemma

Let $\mathbf{Q} \in \mathbb{R}^{d \times d}$ be a symmetric matrix. The linear map $\mathbf{Q} : \mathbb{R}^d \to \mathbb{R}^d$, $\mathbf{w} \mapsto \mathbf{Q}\mathbf{w}$ is a contraction w.r.t. $\|\cdot\|_2$ with constant $\kappa < 1$ if and only if its largest eigenvalue $\lambda_{\max}(\mathbf{Q})$ is bounded by $|\lambda_{\max}(\mathbf{Q})| < 1$; then, $\kappa = |\lambda_{\max}(\mathbf{Q})|$.

Proof.

Via eigenvalue decomposition of $\mathbf{Q}$, as $\mathbf{Q}$ is a symmetric matrix and hence diagonalizable. (Or by next Lemma.) $\square$

- *Affine mapings:* $\mathcal{F}(\mathbf{w}) = \mathbf{Q}\mathbf{w} + \mathbf{b}$ is a contraction with $\kappa < 1$ iff the linear map $\mathbf{Q}$ is a contraction with $\kappa < 1$, as

$$\|\mathcal{F}(\mathbf{w}) - \mathcal{F}(\mathbf{w}')\| = \|\mathbf{Q}\mathbf{w} + \mathbf{b} - (\mathbf{Q}\mathbf{w}' + \mathbf{b})\| = \|\mathbf{Q}\mathbf{w} - \mathbf{Q}\mathbf{w}'\|$$

Contractions on $\mathbb{R}^d$ (non-linear; optional)

Lemma

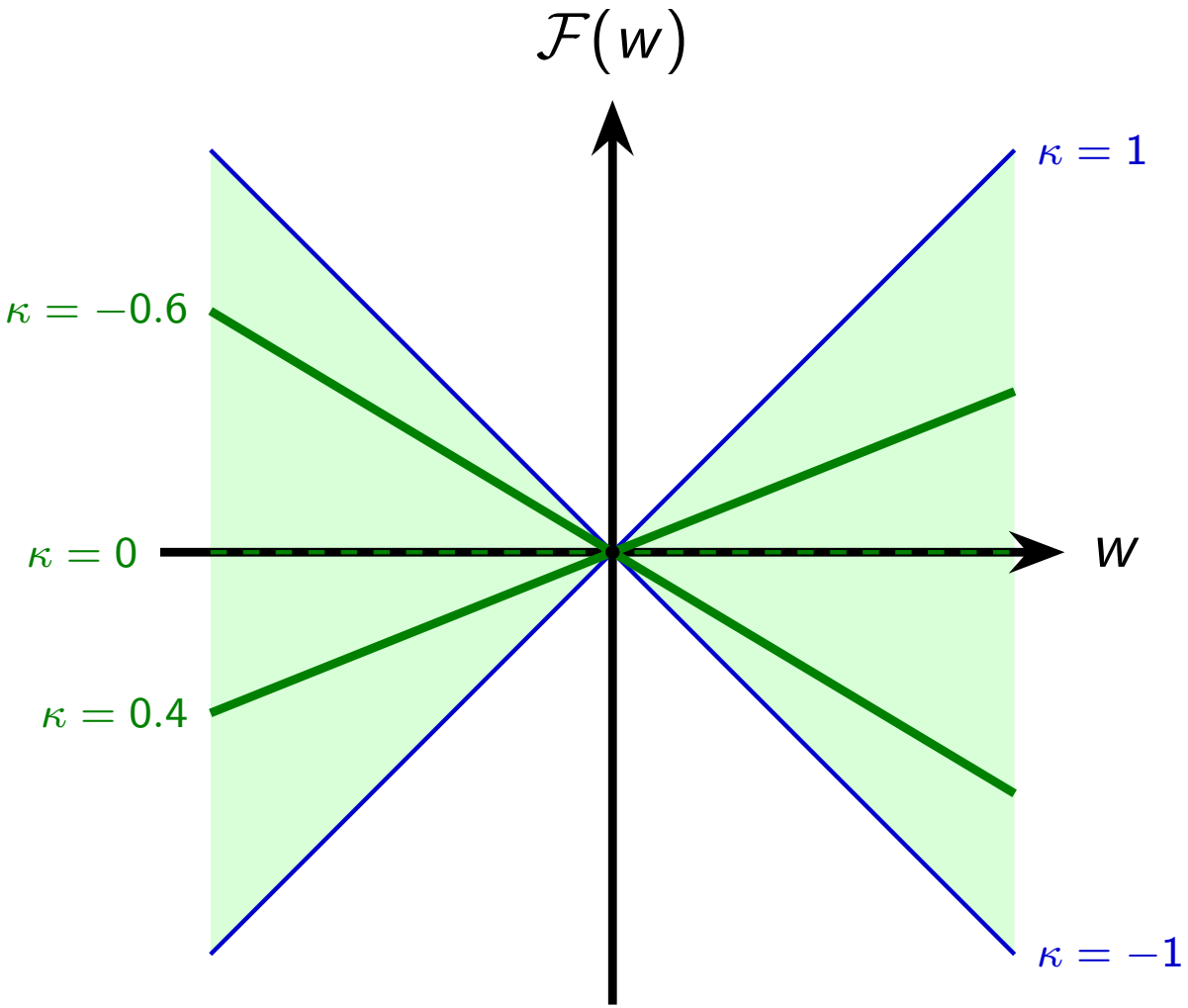
A differentiable function $\mathcal{F} : \mathbb{R}^d \to \mathbb{R}^d$ is a contraction w.r.t. to an arbitrary norm $\|\cdot\|$ with constant $\kappa < 1$ if and only if $\sup_{\mathbf{w} \in \mathbb{R}^d} \|\mathbf{J}_{\mathcal{F}}(\mathbf{w})\| \leq \kappa$, where $\mathbf{J}_f$ denotes the Jacobian of $\mathcal{F}$, and where $\|\cdot\|$ also denotes the induced operator norm on $\mathbb{R}^{d \times d}$.

Proof.

Generalize the one-dimensional case (beyond our scope). $\square$

Illustration: Linear Functions $\mathcal{F}(w) = \kappa \cdot w$

$$|\mathcal{F}(w) - \mathcal{F}(w')| = |\kappa w - \kappa w'| = |\kappa| \cdot |w - w'|$$



contraction

$$\sup_{w \in \mathbb{R}} |\mathcal{F}'(w)| = |\kappa| < 1$$

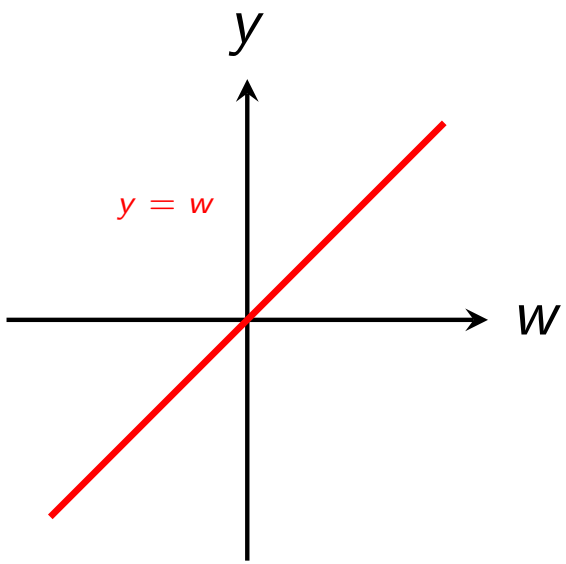
Fixed-Points

A vector $\mathbf{w}^\star \in \mathbb{R}^d$ is fixed point of an operator $\mathcal{F} : \mathbb{R}^d \to \mathbb{R}^d$ if

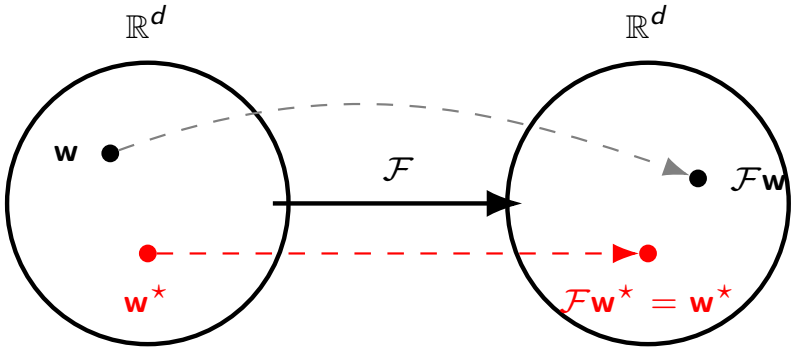
$$\mathcal{F}(\mathbf{w}^\star) = \mathbf{w}^\star.$$

(Analog for $\mathcal{F} : \mathcal{W} \to \mathcal{W}$ for any $\mathcal{W}$, in particular $\mathcal{W} \subset \mathbb{R}^d$.)

Fixed points of $\mathcal{F} : \mathbb{R} \to \mathbb{R}$



Fixed points of $\mathcal{F} : \mathbb{R}^d \to \mathbb{R}^d$



Fixed-Points (ctd.)

Motivation:

- Suitable for root point and fixed point problems:

$$\widehat{\mathcal{F}}(\mathbf{w}^\star) = \mathbf{0} \iff \mathcal{F}(\mathbf{w}^\star) = \mathbf{w}^\star, \quad \text{for} \quad \widehat{\mathcal{F}}(\mathbf{w}) := \mathcal{F}(\mathbf{w}) + \mathbf{w}$$

- ML: solving optimization problems, e.g., ERM instances.

Key Questions:

- Existence and uniqueness of fixed points?
  $\rightsquigarrow$ Banach's fixed-point theorem.
- Practical computation? $\rightsquigarrow$ fixed-point iterations
- Guarantuees? $\rightsquigarrow$ A-priori and a-posteriori error bounds!

Banach's Fixed-Point Theorem

In $(\mathbb{R}^d, \|\cdot\|)$ with some norm $\|\cdot\|$, let $\mathcal{F} : \mathbb{R}^d \to \mathbb{R}^d$ satisfy

$$\|\mathcal{F}(\mathbf{w}) - \mathcal{F}(\mathbf{w}')\| \leq \kappa \cdot \|\mathbf{w} - \mathbf{w}'\| \qquad \forall \mathbf{w}, \mathbf{w}' \in \mathbb{R}^d$$

for some constant $\kappa \in [0, 1)$. Then, $\mathcal{F}$ has a unique fixed point $\mathbf{w}^\star \in \mathbb{R}^d$, i.e., there exists a unique $\mathbf{w}^\star \in \mathbb{R}^d$ such that

$$\mathcal{F}(\mathbf{w}^\star) = \mathbf{w}^\star.$$

Moreover, for any initial $\mathbf{w}^{(0)} \in \mathbb{R}^d$, the fixed-point iteration $\mathbf{w}^{(t)} = \mathcal{F}\left(\mathbf{w}^{(t-1)}\right)$ converges to $\mathbf{w}^\star$, i.e., $\lim_{t\to\infty} \mathbf{w}^{(t)} = \mathbf{w}^\star$.

Every contractive operator $\mathcal{F}$ on $\mathbb{R}^d$ has a unique fixed point, which is obtained as the limit of the fixed-point iteration.

Convergence Speed and Error Guarantees

Assuming Banach's fixed-point theorem, the following holds.

- *A-priori* error bounds (estimates *before* iteration):

$$\text{at } t = 0: \quad \left\|\mathbf{w}^{(t)} - \mathbf{w}^{\star}\right\| \leq \kappa^t \left\|\mathbf{w}^{(0)} - \mathbf{w}^{\star}\right\|,$$

$$\text{at } t = 1: \quad \left\|\mathbf{w}^{(t)} - \mathbf{w}^{\star}\right\| \leq \frac{\kappa^t}{1-\kappa} \left\|\mathbf{w}^{(1)} - \mathbf{w}^{(0)}\right\|.$$

  First upper bound depends on (unknown) fixed point $\mathbf{w}^{\star}$!

- *A-posteriori* error bounds (estimates *during* iteration):

$$\text{at } t+1: \quad \left\|\mathbf{w}^{(t)} - \mathbf{w}^{\star}\right\| \leq \frac{1}{1-\kappa} \left\|\mathbf{w}^{(t+1)} - \mathbf{w}^{(t)}\right\|$$

For more details see [1, Theorem 9.23].

Banach's Fixed-Point Theorem: Remarks

- Can be generalized to complete metric spaces, i.e., metric spaces in which every Cauchy sequence converges.
- Instead of functions $\mathcal{F} : \mathbb{R}^d \to \mathbb{R}^d$, we may also consider $\mathcal{F} : \mathcal{W} \to \mathcal{W}$ with $\mathcal{W} \subset \mathbb{R}^d$, so-called *self-mappings*.
- Note: a function $\mathcal{F} : \mathcal{W} \to \mathcal{U}$ may be a contraction, but may not have a fixed point when the *self-mapping* property is violated, i.e., when we cannot choose $\mathcal{U} = \mathcal{W}$.
- Weakening the contraction property by only assuming

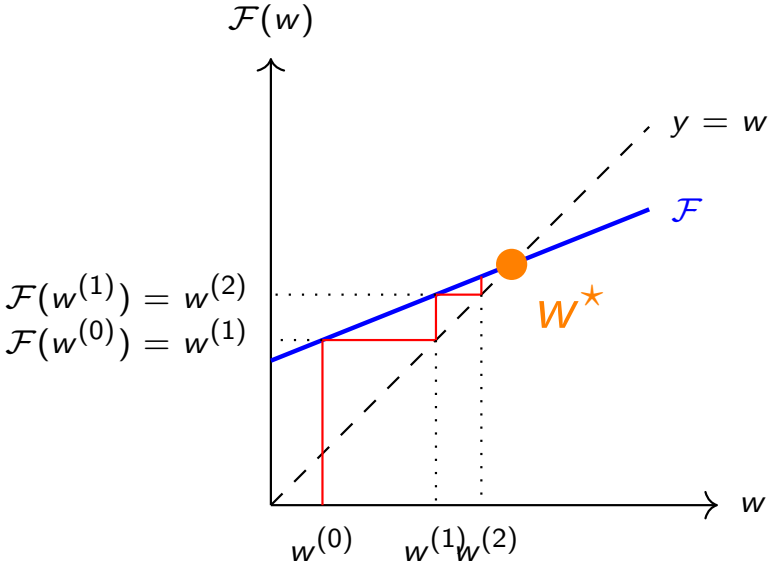
$$\|\mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}'\| < \|\mathbf{w} - \mathbf{w}'\| \qquad \forall \mathbf{w}, \mathbf{w}' \in \mathbb{R}^d$$

  is insufficient to guarantee the existence of a fixed point.

Fixed-Point Iterations and Contractions

Recall the fixed-point iterations $w^{(t+1)} = \mathcal{F}(w^{(t)})$.

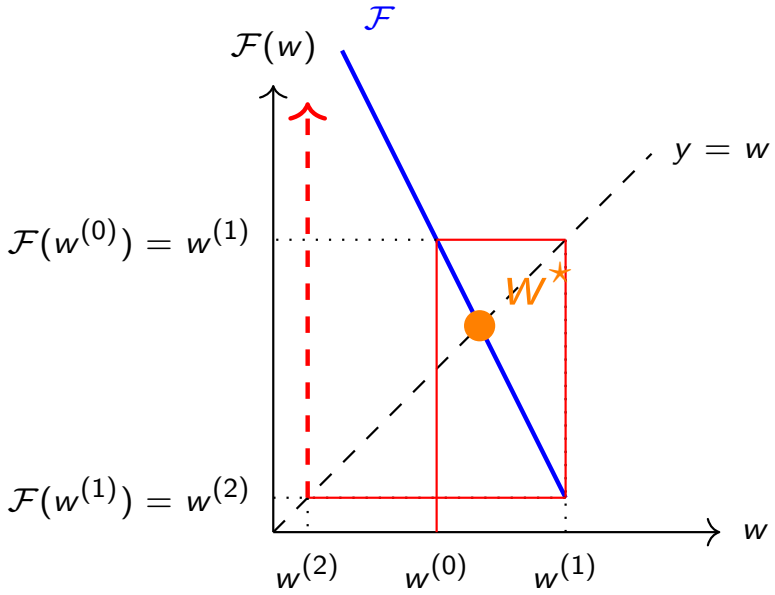
Convergence



$$|\mathcal{F}'(w)| = 0.4 < 1$$

(contraction)

Divergence



$$|\mathcal{F}'(w)| = 2 > 1$$

(expansive)

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(Firmly) Non-Expansive Operators

An operator $\mathcal{F} : \mathbb{R}^d \to \mathbb{R}^d$ is called

- **non-expansive** with respect to some norm $\|\cdot\|$, if

$$\|\mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}'\| \leq \|\mathbf{w} - \mathbf{w}'\| \qquad \forall \mathbf{w}, \mathbf{w}' \in \mathbb{R}^d,$$

- **firmly non-expansive** (recall inner product on $\mathbb{R}^d$), if

$$\|\mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}'\|_2^2 \leq \langle \mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}', \mathbf{w} - \mathbf{w}' \rangle \qquad \forall \mathbf{w}, \mathbf{w}' \in \mathbb{R}^d.$$

$\mathcal{F}$ firmly non-expansive $\Rightarrow$ $\mathcal{F}$ non-expansive for ℓ_2-norm

$$\begin{aligned} \|\mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}'\|_2^{\cancel{2}} &\leq \langle \mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}', \mathbf{w} - \mathbf{w}' \rangle \\ &\leq \cancel{\|\mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}'\|_2} \cdot \|\mathbf{w} - \mathbf{w}'\|_2 \end{aligned}$$

(Cauchy-Schwarz inequality)

Firmly Non-expansive and Averaged Operators

$\mathcal{F}$ is firmly non-expansive operator

$$\iff$$

$\mathcal{F} = \frac{1}{2}\,\mathrm{Id} + \frac{1}{2}\widehat{\mathcal{F}}$ for some non-expansive $\widehat{\mathcal{F}}$

[For more details and a proof: [2, Proposition 4.35].]

Characterization of linear functions:

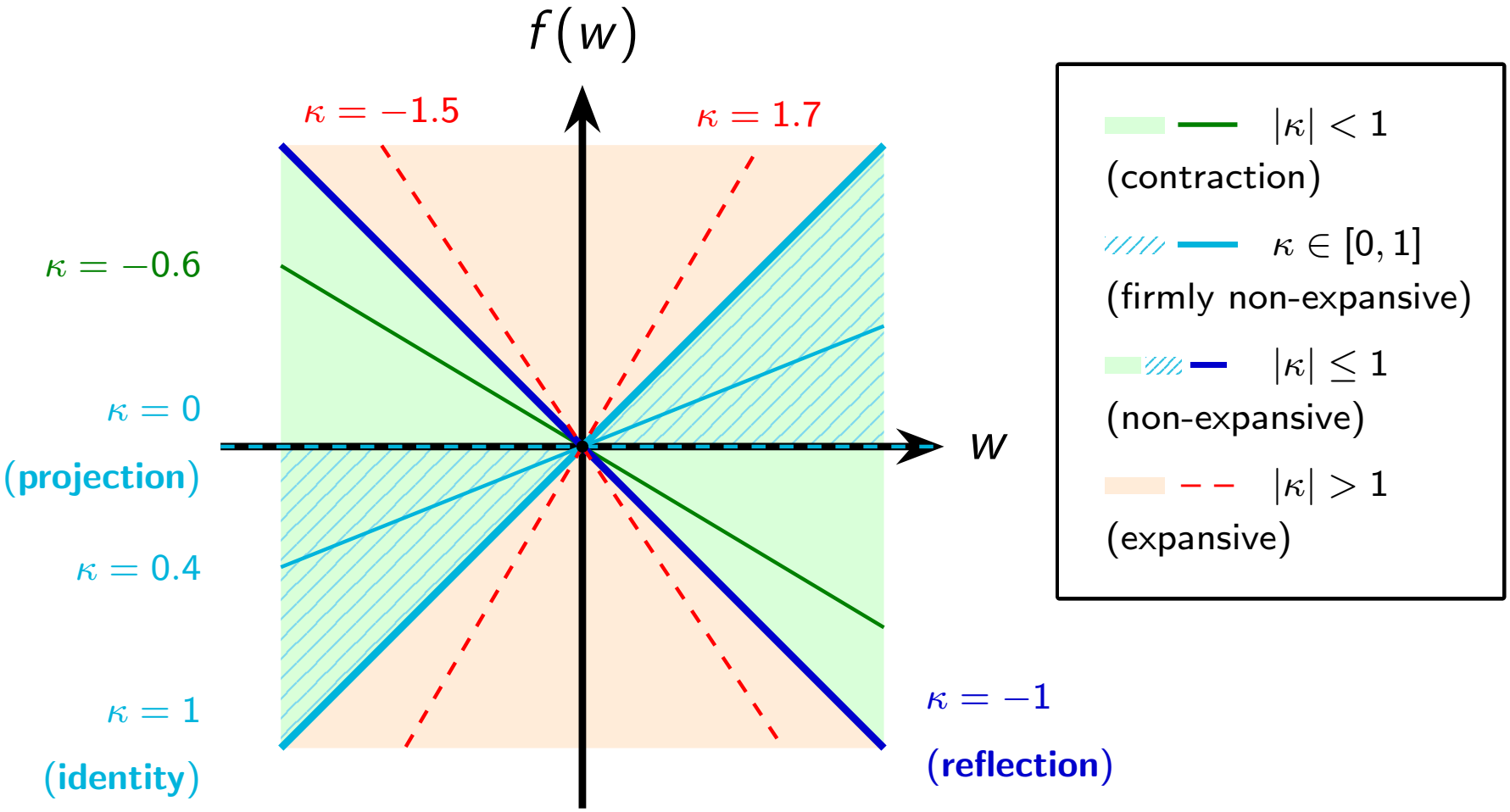
$\mathcal{F} : \mathbb{R}^d \to \mathbb{R}^d, \mathbf{w} \mapsto \kappa\mathbf{w}$ is firmly non-expansive $\Leftrightarrow$ $\kappa \in [0, 1]$.

- *Averaging* $\kappa\mathbf{w} = \frac{1}{2}\mathbf{w} + \frac{1}{2}\widehat{\kappa}\mathbf{w}$ with $|\widehat{\kappa}| \leq 1 \rightsquigarrow \kappa \in [0, 1]$.
- *Manually:* Using $\langle \mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}', \mathbf{w} - \mathbf{w}' \rangle \geq \|\mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}'\|_2^2$:

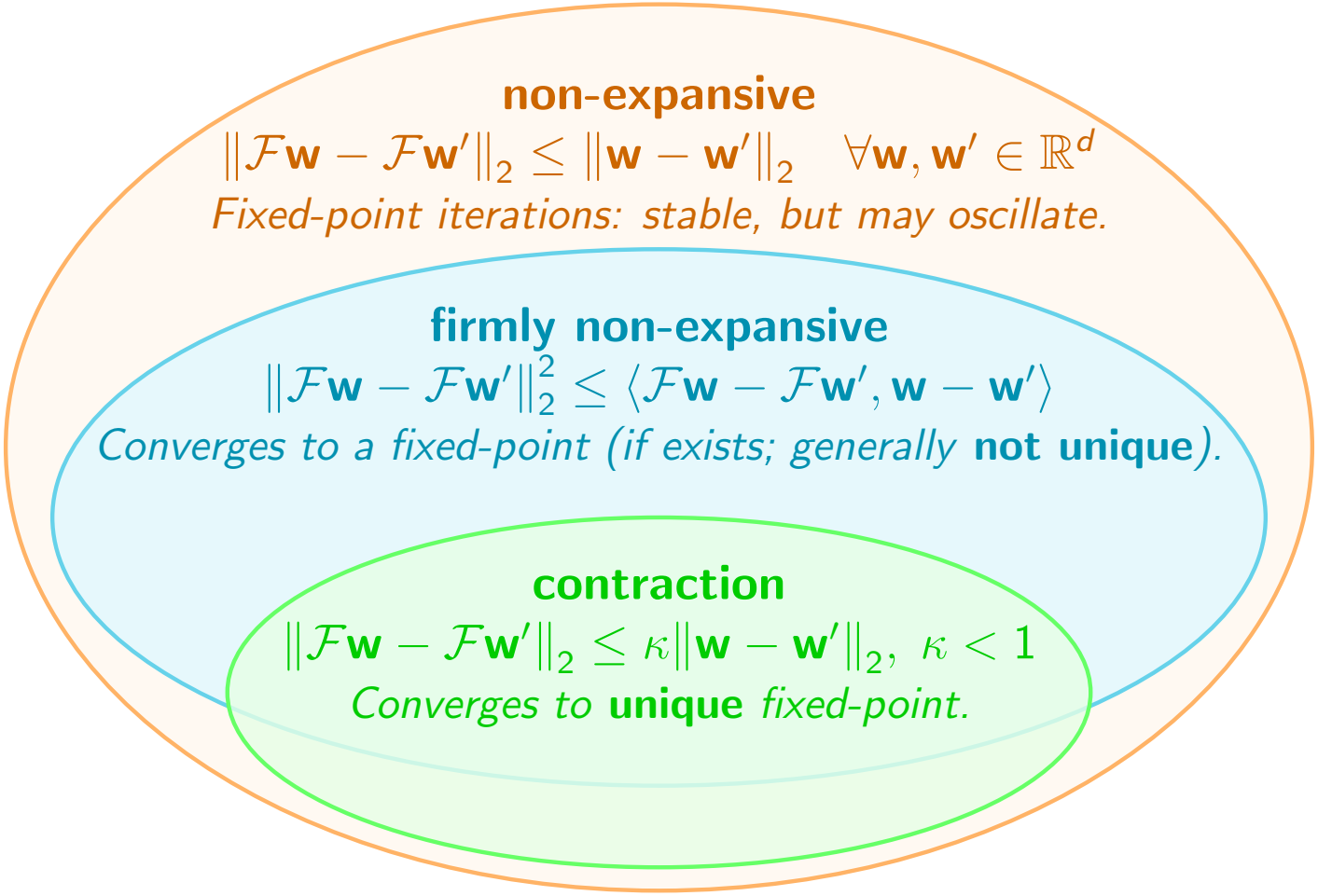
$$\begin{aligned} &\langle \kappa(\mathbf{w} - \mathbf{w}'), \mathbf{w} - \mathbf{w}' \rangle \geq \|\kappa(\mathbf{w} - \mathbf{w}')\|_2^2 \\ \iff &\kappa\|\mathbf{w} - \mathbf{w}'\|_2^2 \geq \kappa^2\|\mathbf{w} - \mathbf{w}'\|_2^2 \\ \iff &\kappa \geq \kappa^2 \iff \kappa(1 - \kappa) \geq 0 \iff \kappa \in [0, 1]. \end{aligned}$$

Illustration with Linear Functions $\mathcal{F}(w) = \kappa \cdot w$

$$|\mathcal{F}(w) - \mathcal{F}(w')| = |\kappa w - \kappa w'| = |\kappa| \cdot |w - w'|$$



Hierarchy of Fixed-Point Operators (for ℓ_2-norm)



Venn diagram illustrating relations between operator classes via set inclusions; convergence of their fixed-point iteration

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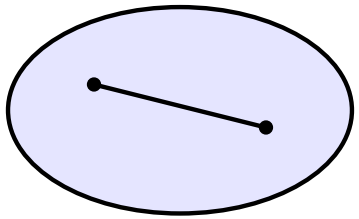
Convexity

(1) Convex set $\mathcal{C} \subset \mathbb{R}^d$: Line segment between any $\mathbf{x}, \mathbf{y} \in \mathcal{C}$ remains in $\mathcal{C}$.

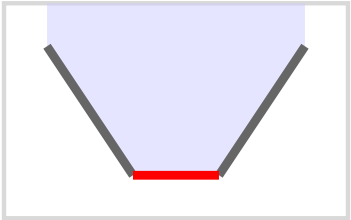
$$\alpha\mathbf{x} + (1-\alpha)\mathbf{y} \in \mathcal{C} \quad \forall \alpha \in [0,1]$$



(1) Convex Set

(2) Convex function f: The epigraph is a convex set, i.e., for $\alpha \in [0,1]$:

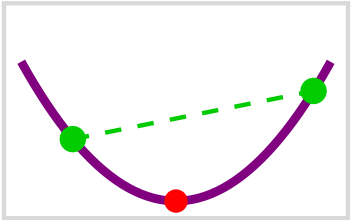
$$f(\alpha\mathbf{x}+(1-\alpha)\mathbf{y}) \leq \alpha f(\mathbf{x})+(1-\alpha)f(\mathbf{y})$$



(2) Convex Function

(3) Strictly convex function f: Strict inequality for $\mathbf{x} \neq \mathbf{y}$ and $\alpha \in (0,1)$:

$$f(\alpha\mathbf{x} + (1-\alpha)\mathbf{y}) < \alpha f(\mathbf{x}) + (1-\alpha)f(\mathbf{y})$$



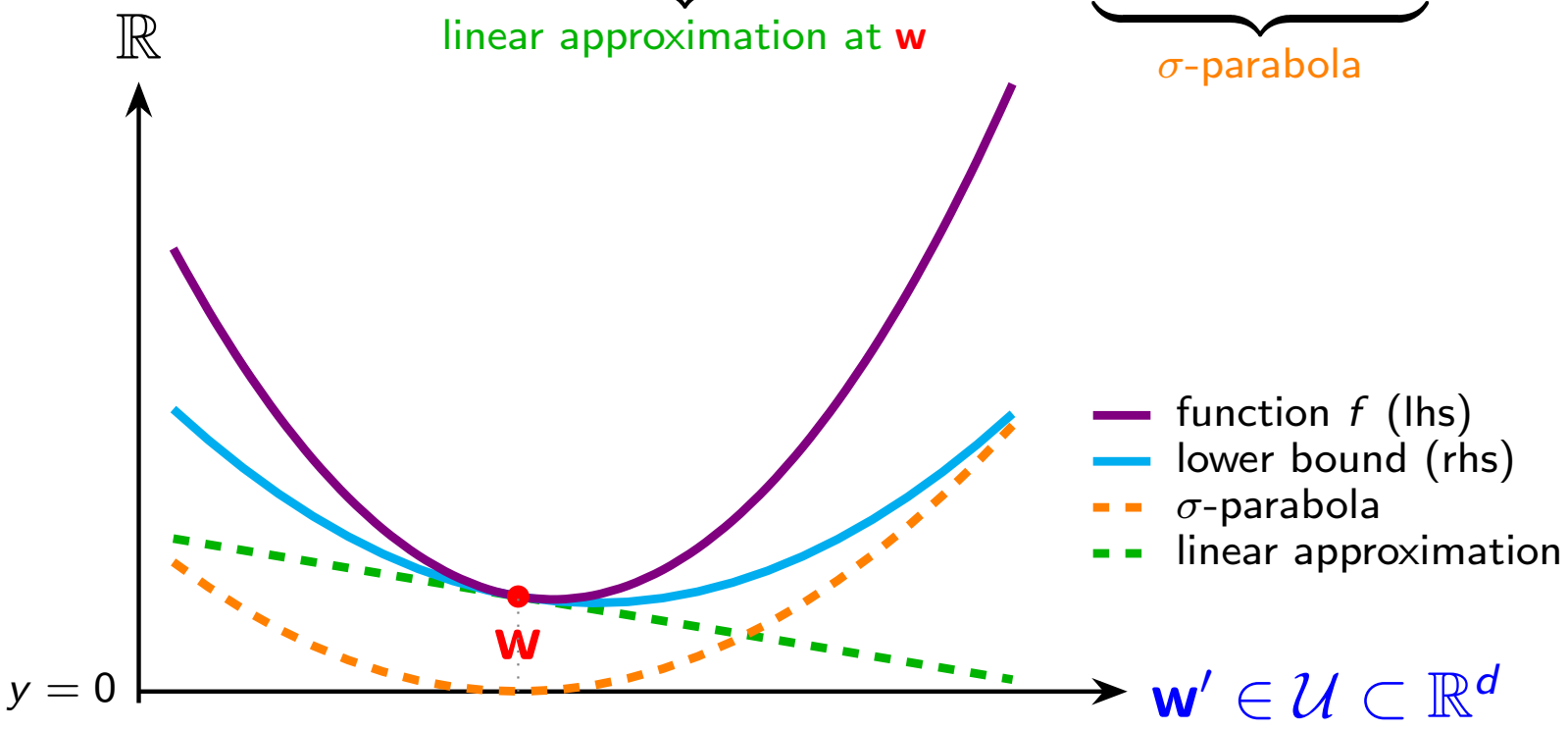
(3) Strictly Convex

(Unique) minimizer exist(s) for (strictly) convex functions.

Strong Convexity

A differentiable function $f : \mathcal{U} \to \mathbb{R}$ with domain $\mathcal{U} \subset \mathbb{R}^d$ is **strongly convex** with coefficient $\sigma > 0$ if for all $\mathbf{w}, \mathbf{w}' \in \mathcal{U}$

$$f(\mathbf{w}') \geq \underbrace{f(\mathbf{w}) + \nabla f(\mathbf{w})^\top (\mathbf{w}' - \mathbf{w})}_{\text{linear approximation at } \mathbf{w}} + \underbrace{\frac{\sigma}{2} \|\mathbf{w}' - \mathbf{w}\|_2^2}_{\sigma\text{-parabola}}.$$

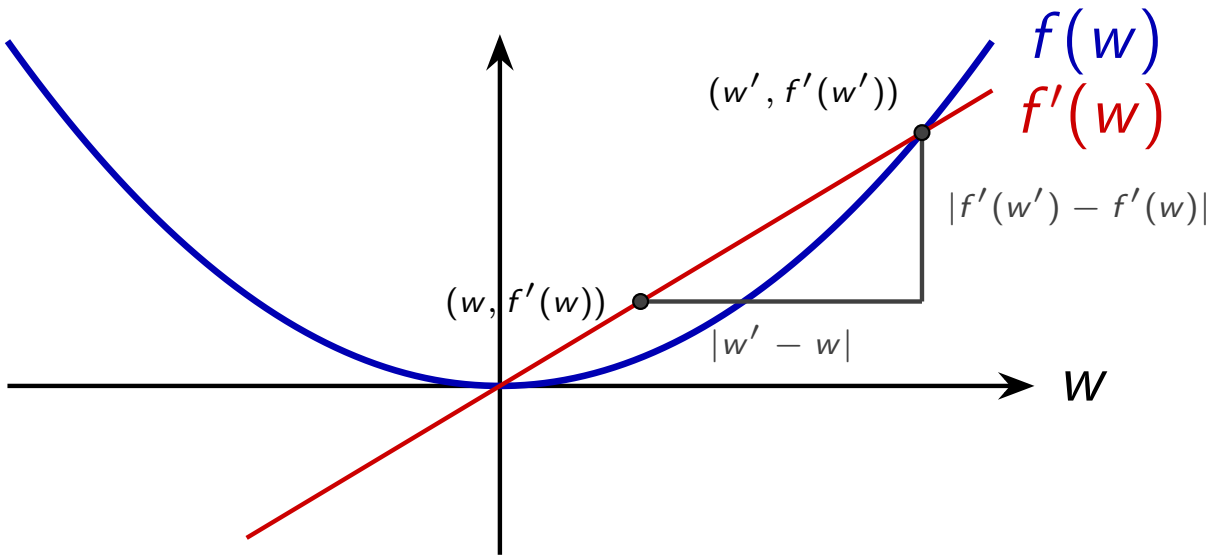


$$\boxed{\begin{array}{ccccc} \sigma\text{-strongly convex} & \Longrightarrow & f \text{ strictly convex} & \Longrightarrow & f \text{ convex} \\ \sigma\text{-strongly convex} & \not\Longleftarrow & f \text{ strictly convex} & \not\Longleftarrow & f \text{ convex} \end{array}}$$

Smoothness

A smooth function f is referred to as β-**smooth** if its gradient ∇f is Lipschitz-continuous with constant $\beta > 0$, i.e., when

$$\|\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')\|_2 \leq \beta \|\mathbf{w} - \mathbf{w}'\|_2 \quad \forall \mathbf{w}, \mathbf{w}' \in \mathbb{R}^d.$$

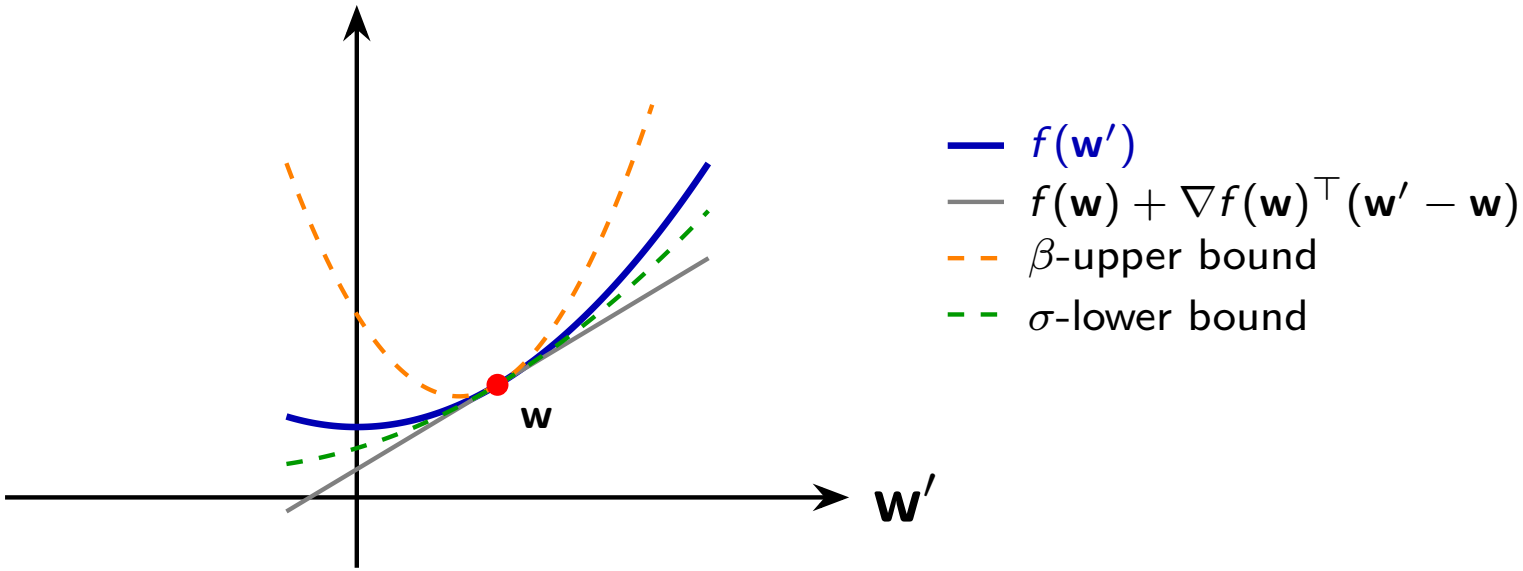


$f(w) = \frac{\beta}{2} w^2$ is β-smooth, as $f'(w) = \beta w$ has a slope of β.

Quadratic Upper (and Lower) Bound

Any β-smooth function $f : \mathbb{R}^d \to \mathbb{R}$ satisfies [4, Lemma 3.4]

$$f(\mathbf{w}') \leq f(\mathbf{w}) + \nabla f(\mathbf{w})^\top (\mathbf{w}' - \mathbf{w}) + \frac{\beta}{2}\|\mathbf{w}' - \mathbf{w}\|_2^2 \quad \forall \mathbf{w}, \mathbf{w}' \in \mathbb{R}^d.$$



β-smoothness majorizes the function f by a parabola. If f is also σ-strongly convex, it is *sandwiched* in-between two parabolas, with $\Delta\mathbf{w} = \mathbf{w}' - \mathbf{w}$ it holds:

$$f(\mathbf{w}) + \nabla f(\mathbf{w})^\top \Delta\mathbf{w} + \frac{\sigma}{2}\|\Delta\mathbf{w}\|^2 \leq f(\mathbf{w}') \leq f(\mathbf{w}) + \nabla f(\mathbf{w})^\top \Delta\mathbf{w} + \frac{\beta}{2}\|\Delta\mathbf{w}\|^2$$

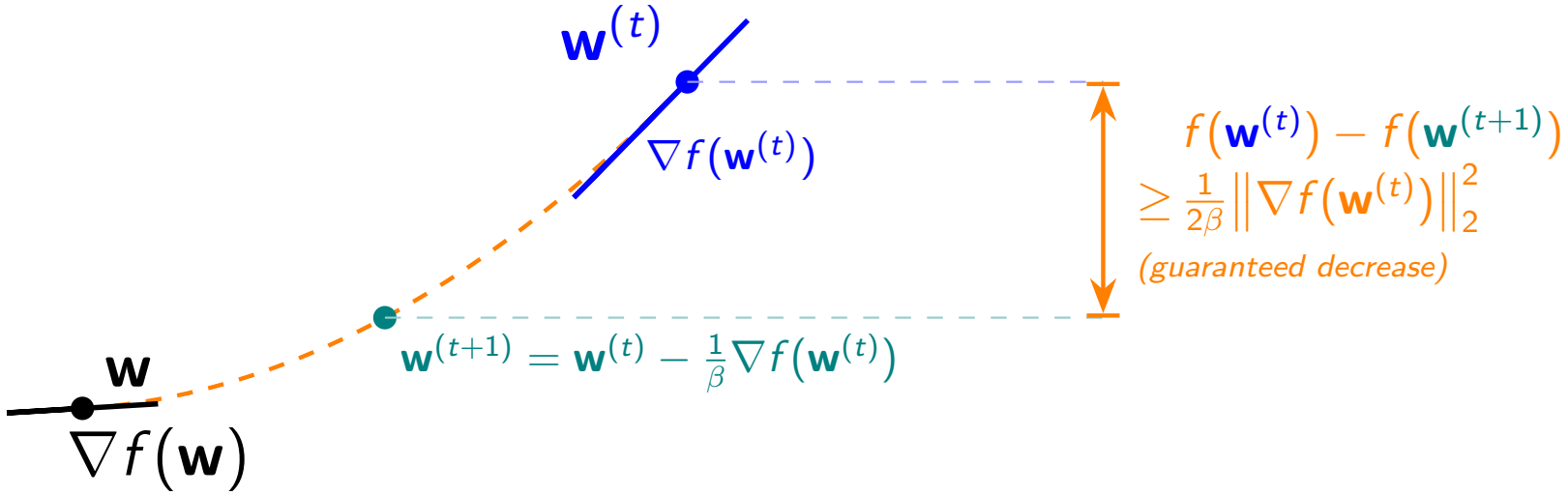
Descent Lemma and Step Size

For β-smooth f, upper bound at t-th step of GD with $\mathbf{w}^{(t)}$ is

$$f(\mathbf{w}) \leq f(\mathbf{w}^{(t)}) + \nabla f(\mathbf{w}^{(t)})^\top (\mathbf{w} - \mathbf{w}^{(t)}) + \frac{\beta}{2} \left\| \mathbf{w} - \mathbf{w}^{(t)} \right\|_2^2.$$

Minimizing upper bound with respect to $\mathbf{w}$ yields gradient step

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \frac{1}{\beta} \nabla f(\mathbf{w}^{(t)}).$$



β-smoothness guarantees decrease for each gradient step ($\eta = \frac{1}{\beta}$), but (how quickly) do we converge to the optimum?

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Gradient Descent and Proximal Operators

Zero-Gradient Condition and Gradient Descent

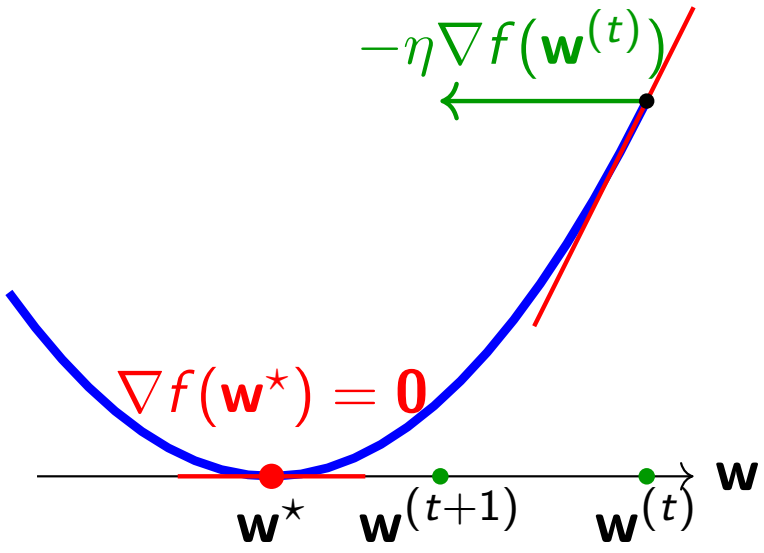
- For $f : \mathbb{R}^d \to \mathbb{R}$, consider unconstrained optimization problem

$$\min_{\mathbf{w} \in \mathbb{R}^d} f(\mathbf{w}).$$

- For a differentiable and convex f, $\mathbf{w}^\star$ is a global minimizer, if and only if the zero-gradient condition is satisfied, i.e., when

$$\nabla f\left(\mathbf{w}^\star\right) = \mathbf{0}$$

- If $\nabla f(\mathbf{w}^\star) = \mathbf{0}$ is hard to solve explicitly, we use iterative updates, with initialization $\mathbf{w}^{(0)}$ and step size $\eta > 0$.



$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \nabla f(\mathbf{w}^{(t)})$$

A Fixed-Point View of Gradient Descent

- **Gradient step operator** $\mathcal{T}^{(f,\eta)}$ for smooth f and $\eta > 0$:

$$\mathcal{T}^{(f,\eta)} : \mathbb{R}^d \to \mathbb{R}^d, \quad \mathbf{w} \mapsto \mathbf{w} - \eta \nabla f(\mathbf{w})$$

- Minimizing f via the zero-gradient condition is equivalent to solving a fixed-point equation for any step size $\eta > 0$:

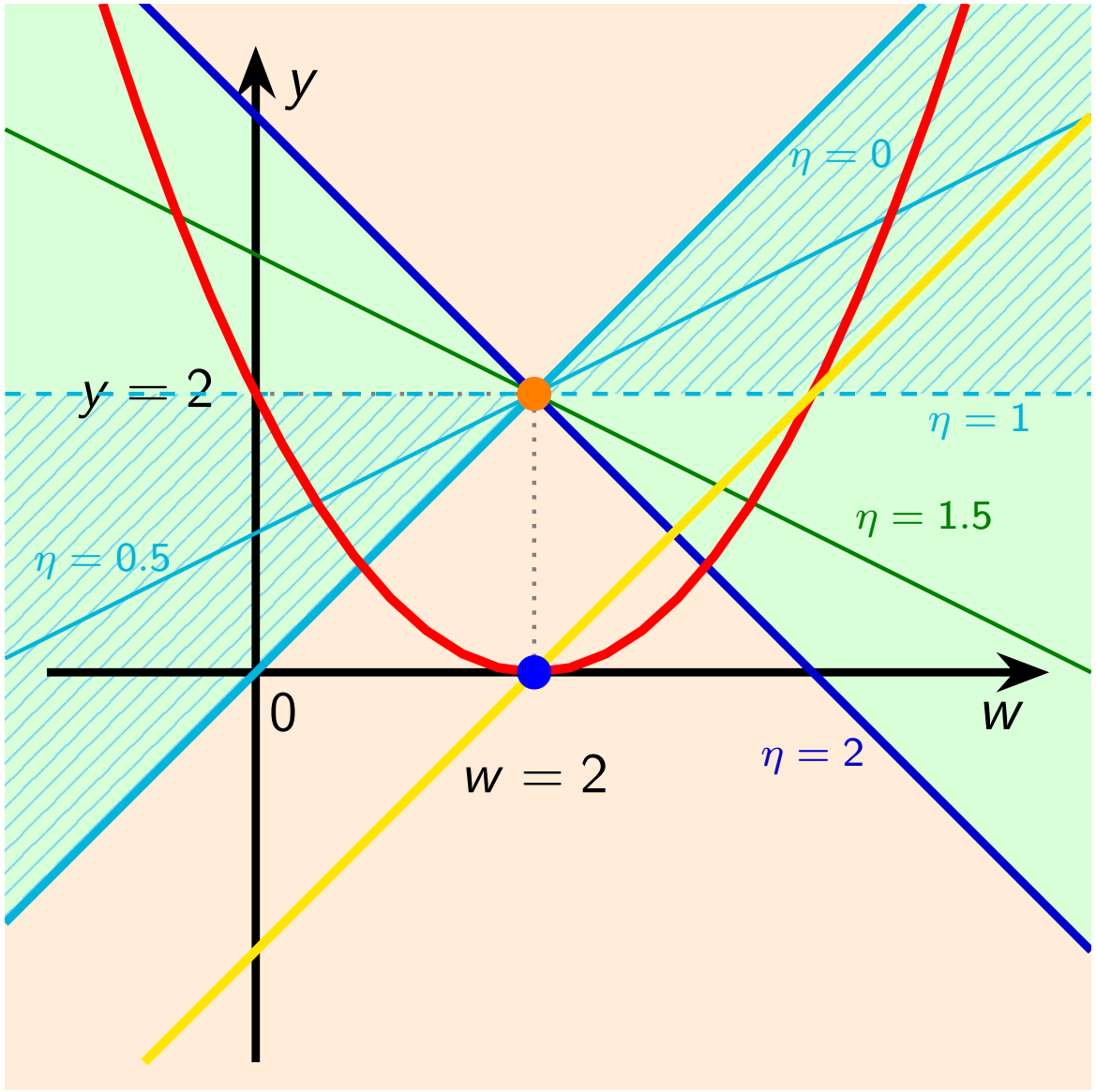
$$\begin{aligned} \mathbf{w}^\star \in \arg\min_{\mathbf{w} \in \mathbb{R}^d} f(\mathbf{w}) &\iff \nabla f\left(\mathbf{w}^\star\right) = \mathbf{0} \\ &\iff \mathbf{w}^\star - \eta \nabla f(\mathbf{w}^\star) = \mathbf{w}^\star \\ &\iff \mathcal{T}^{(f,\eta)}(\mathbf{w}^\star) = \mathbf{w}^\star \end{aligned}$$

GD with step size η is the fixed-point iteration of $\mathcal{T}^{(f,\eta)}$:

$$\mathbf{w}^{(t+1)} = \mathcal{T}^{(f,\eta)}(\mathbf{w}^{(t)}).$$

We can apply Banach's fixed-point theorem!

Illustration: Gradient Descent and Fixed-Points



Function and Derivative

$f(w) = \frac{1}{2}(w-2)^2$

$\nabla f(w) = w - 2$

Gradient Step Op. $\mathcal{T}^{(f,\eta)}$

$\mathcal{T}(w) = (1-\eta)w + 2\eta$

Operator Properties

$\eta \in (0,2)$ (contraction)

$\eta \in [0,1]$ (firmly non-expansive)

$\eta \in [0,2]$ (non-expansive)

$\eta \notin [0,2]$ (expansive)

Important Points

- fixed-point
- root of ∇f (zero-gradient condition)

Properties of the Gradient Step Operator

Prop. of $\mathcal{T}^{(f,\eta)}(\mathbf{w}) = \mathbf{w} - \eta\nabla f(\mathbf{w})$ for β-smooth convex f:

- $\mathcal{T}^{(f,\eta)}$ is non-expansive operator if $\eta \in (0, 2/\beta]$.
- $\mathcal{T}^{(f,\eta)}$ is firmly non-expansive operator if $\eta \in (0, 1/\beta]$.
- If f is additionally σ-strongly convex, then $\mathcal{T}^{(f,\eta)}$ is a contractive operator for any $\eta \in (0, 2/\beta)$.

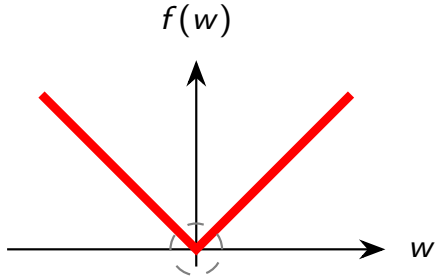
$$\left\|\mathcal{T}^{(f,\eta)}(\mathbf{w}) - \mathcal{T}^{(f,\eta)}(\mathbf{w}')\right\|_2^2 \leq \left(1 - \eta\sigma^2\left(\frac{2}{\beta} - \eta\right)\right)\|\mathbf{w} - \mathbf{w}'\|_2^2.$$

Indeed, $\kappa(\eta) := 1 - \eta\sigma^2(2/\beta - \eta) < 1$ for $\eta \in (0, 2/\beta)$, and the operator $\mathcal{T}^{(f,\eta)}$ is a contraction in this case; note that $\kappa(\eta)$ becomes minimal for $\eta = 1/\beta$.

Proof via [4, Lemma 3.11]; see bonus slides in the end.

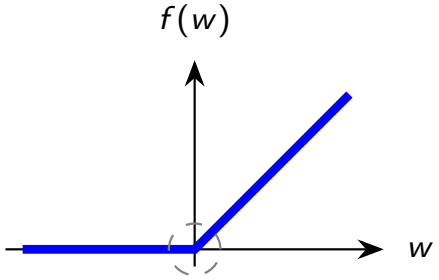
Non-Differentiable Optimization Problems

Lasso (ℓ_1-regularization)



- $f(w) = \lambda|w|$ for $\lambda > 0$, non-differentiable in 0.
- Promotes *sparsity* by pushing weights to zero.

Neural Networks (ReLU)

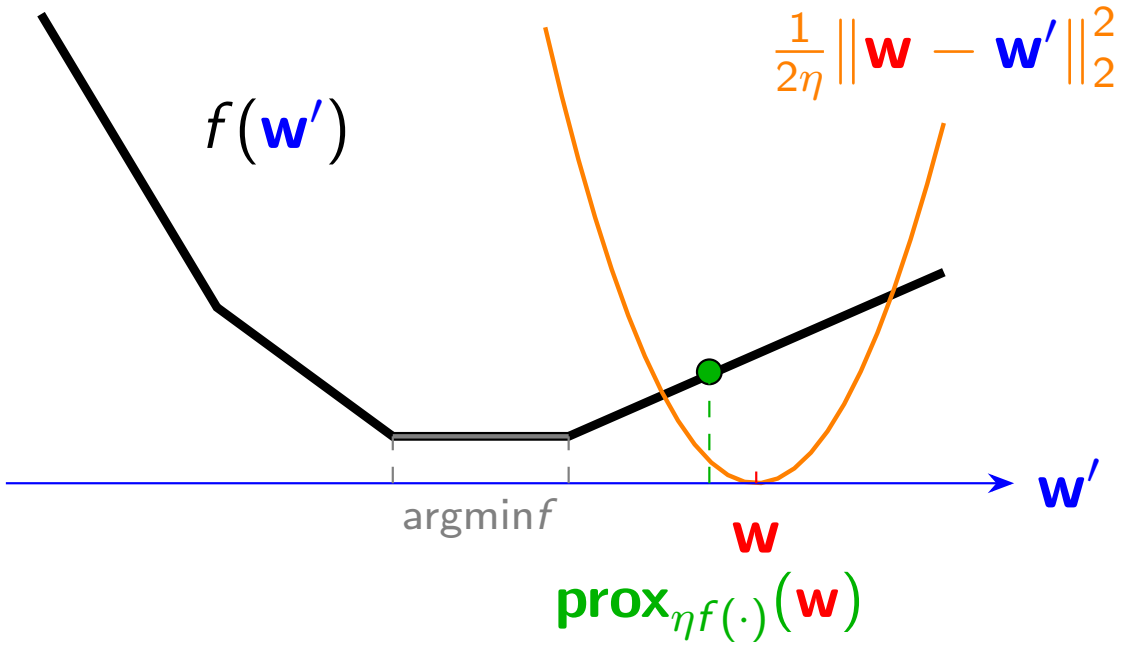


- $f(w) = \max(0, w)$, non-differentiable in 0.
- Standard activation function for ANNs.

Proximal Operators

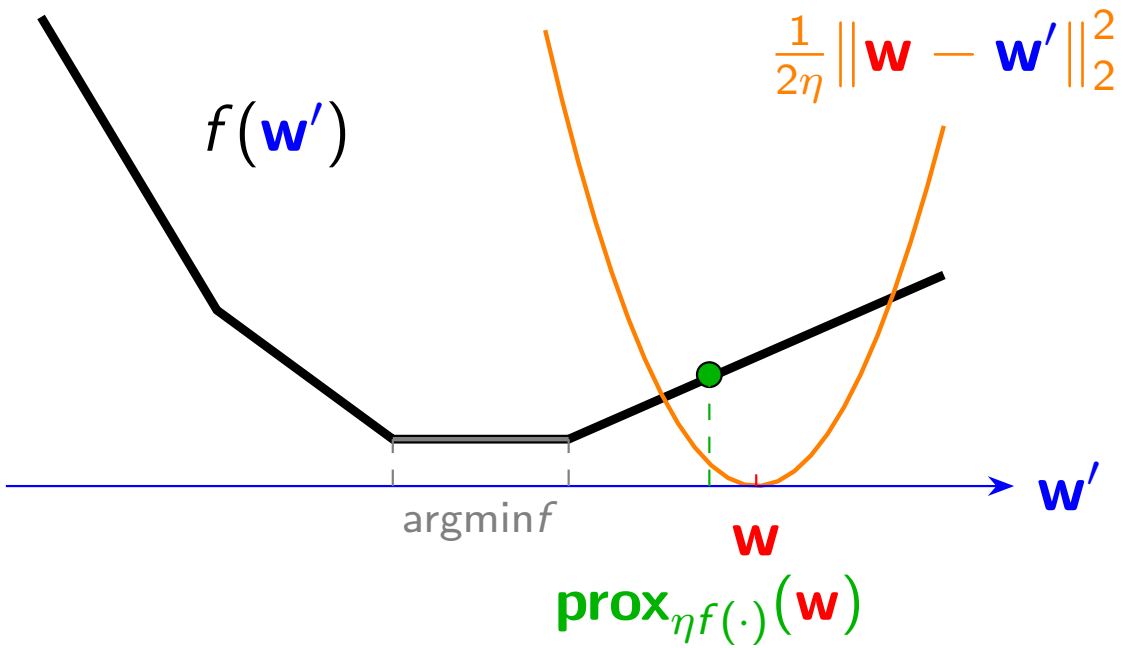
For a continuous convex function $f : \mathbb{R}^d \to \mathbb{R}$, and any $\eta > 0$, we define the **proximal operator** $\mathbf{prox}_{\eta f(\cdot)}(\mathbf{w})$ of f as

$$\mathbf{prox}_{\eta f(\cdot)}(\mathbf{w}) := \arg\min_{\mathbf{w}' \in \mathbb{R}^d} \left[\underbrace{f(\mathbf{w}')}_{\text{convex}} + \underbrace{\frac{1}{2\eta}\|\mathbf{w} - \mathbf{w}'\|_2^2}_{\text{strictly convex}} \right].$$



Proximal Operators (ctd.)

$$\mathbf{prox}_{\eta f(\cdot)}(\mathbf{w}) := \underset{\mathbf{w}' \in \mathbb{R}^d}{\arg\min} \left[\underbrace{f(\mathbf{w}')}_{\text{convex}} + \underbrace{\frac{1}{2\eta}\|\mathbf{w} - \mathbf{w}'\|_2^2}_{\text{strictly convex}} \right].$$



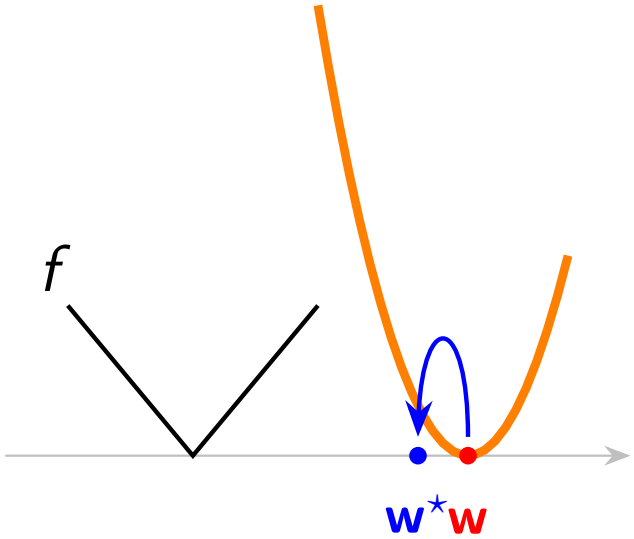
- Objective always strictly convex (*unique minimizer*)!

Remarks on the Step Size η

- Minimize f while staying close to the input $\mathbf{w} \in \mathbb{R}^d$.
- Justifying the notation - scaling of f and penalty:

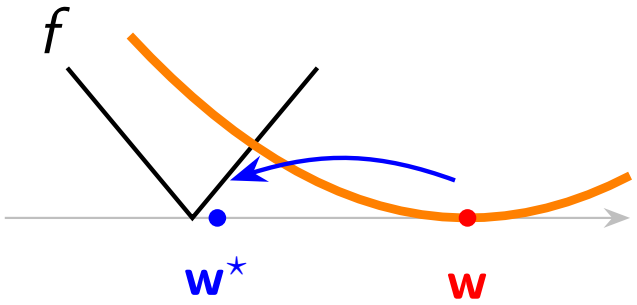
$$\begin{aligned}\mathbf{prox}_{\eta f}(\mathbf{w}) &= \arg\min_{\mathbf{w}' \in \mathbb{R}^d} \left[\eta f(\mathbf{w}') + \tfrac{1}{2}\|\mathbf{w} - \mathbf{w}'\|_2^2\right] \\ &= \arg\min_{\mathbf{w}' \in \mathbb{R}^d} \left[f(\mathbf{w}') + \tfrac{1}{2\eta}\|\mathbf{w}\|_\mathbf{w}'2^2\right] =: \mathbf{w}^\star.\end{aligned}$$

small η (strong penalty)



penalty dominates

large η (weak penalty)



function f dominates

Proximal Operator as a Generalized Gradient Step

- For linear approx. $\tilde{f}(\mathbf{w}') = f(\mathbf{w}) + \nabla f(\mathbf{w})^T(\mathbf{w}' - \mathbf{w})$:

$$\mathbf{prox}_{\eta\tilde{f}(\cdot)}(\mathbf{w}) = \underset{\mathbf{w}' \in \mathbb{R}^d}{\arg\min} \left[f(\mathbf{w}) + \nabla f(\mathbf{w})^T(\mathbf{w}' - \mathbf{w}) + \frac{1}{2\eta}\|\mathbf{w} - \mathbf{w}'\|_2^2 \right]$$

- The zero-gradient condition (gradient w.r.t. $\mathbf{w}'$) yields

$$\nabla f(\mathbf{w}) + \frac{1}{\eta}(\mathbf{w}' - \mathbf{w}) = 0.$$

- Equivalent to standard gradient step with step size $\eta > 0$:

$$\mathbf{w}' = \mathbf{w} - \eta \nabla f(\mathbf{w}).$$

The proximal operator generalizes gradient-based methods. If we replace the function f with its linear approximation $\tilde{f}$ at $\mathbf{w}$, the proximal operator with scaling factor η is equivalent to a gradient step step of f with learning rate η.

Proximal Steps

- Generalization of ∇-step even to non-differentiable f:

$$\mathbf{w}^{(t+1)} = \mathbf{prox}_{\eta f(\cdot)}(\mathbf{w}^{(t)}),$$

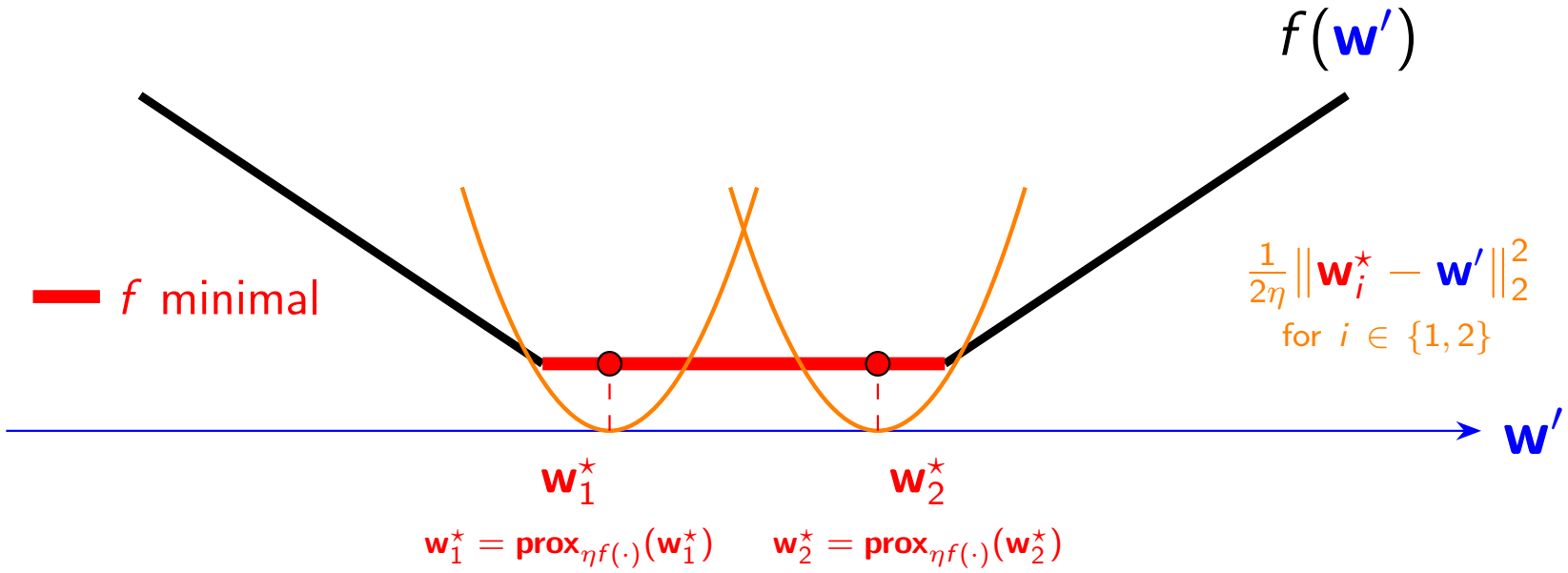
  which defines a **proximal step** with initial $\mathbf{w}^{(0)} \in \mathbb{R}^d$.
- Converges (when?) to a minimizer of function f?
- Analyze proximal methods from a fixed point view!

$$\mathbf{w}^\star = \mathbf{prox}_{\eta f(\cdot)}(\mathbf{w}^\star)$$

A Fixed-Point View of Proximal Operators

For a continuous convex function f, the set of its minimizers coincides with the set of fixed points of its proximal operator.

$$\mathbf{w}^\star \in \arg\min_{\mathbf{w}' \in \mathbb{R}^d} f(\mathbf{w}') \iff \mathbf{w}^\star = \mathbf{prox}_{\eta f(\cdot)}(\mathbf{w}^\star) \quad \forall \eta > 0$$



For more details and a formal proof: [3, Section 2.3].

Properties of Proximal Operators

- Any proximal operator is also a firmly non-expansive operator [2, Prop 12.28].
- Proximal operators can be even contractions under additional assumptions. If f is σ-strongly convex and

$$\kappa = \frac{1}{1 + \eta\sigma} \in (0, 1),$$

then the proximal operator of f is a contractive operator with contraction factor κ [2, Prop 23.13].

Proximal Operator of the Absolute Value Function

Let $f(w) = |w|$. To find $\mathbf{prox}_{\eta f(\cdot)}(w)$, minimize the objective

$$J(w') = |w'| + \frac{1}{2\eta}(w - w')^2.$$

Both f, J are non-differentiable at $w = 0$. Consider the cases:

- **Case 1:** $w' > 0$. Then $|w'| = w'$.

$$\frac{d}{dw'}\left[w' + \frac{1}{2\eta}(w - w')^2\right] = 1 - \frac{1}{\eta}(w - w') = 0$$

$$\implies w' = w - \eta \quad \text{(Valid if } w > \eta\text{)}$$

- **Case 2:** $w' < 0$. Then $|w'| = -w'$.

$$\frac{d}{dw'}\left[-w' + \frac{1}{2\eta}(w - w')^2\right] = -1 - \frac{1}{\eta}(w - w') = 0$$

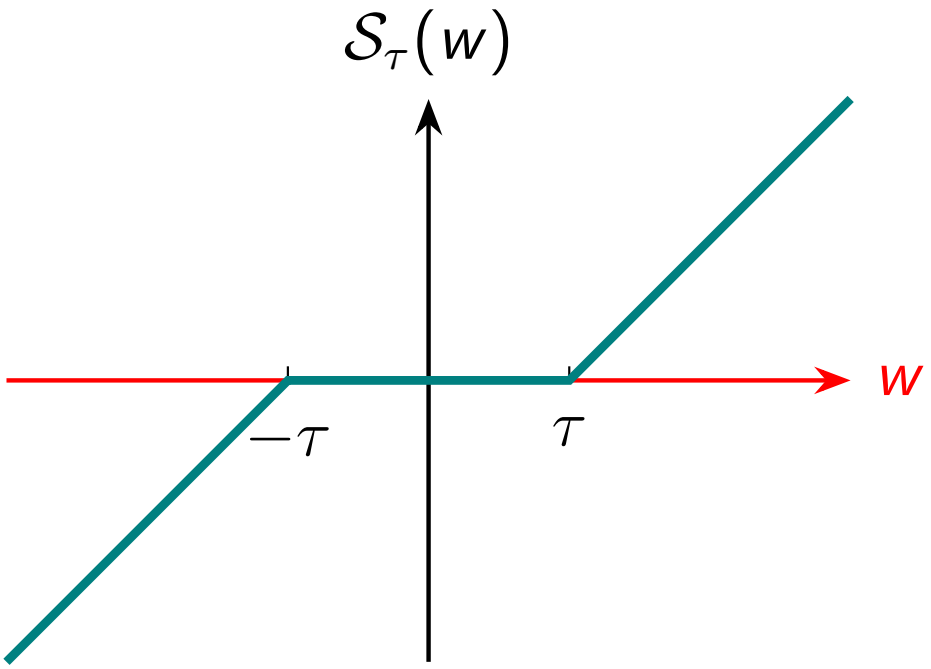
$$\implies w' = w + \eta \quad \text{(Valid if } w < -\eta\text{)}$$

- **Case 3:** If $|w| \leq \eta$, the quadratic penalty is "too weak" to pull the minimizer away from $w' = 0$.

The Soft-Thresholding Function

The proximal operator is $\mathbf{prox}_{\eta|\cdot|}(w) = \mathcal{S}_\eta(w)$, where the **soft-thresholding** function $\mathcal{S}_\tau(w)$ is defined as (for $\tau \geq 0$)

$$\mathcal{S}_\tau(w) = \text{sign}(w) \cdot \max(|w| - \tau, 0) = \begin{cases} w - \tau & w > \tau \\ 0 & |w| \leq \tau \\ w + \tau & w < -\tau \end{cases}$$



More generally, this is useful in the context of ℓ_1-regularization, such as for the Lasso.

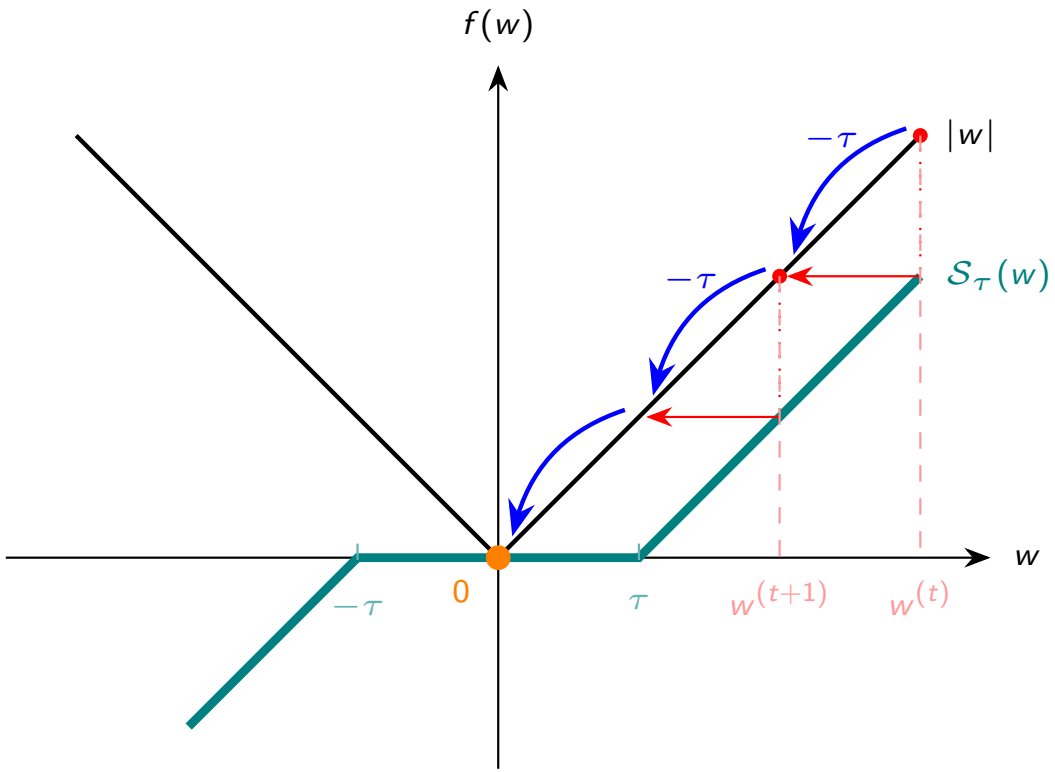
Convergence of Soft-Thresholding Iterations

For any $w^{(0)}$, the iteration $w^{(t+1)} = \mathcal{S}_\tau(w^{(t)})$ converges to 0 by taking τ-steps toward the origin until the optimum is reached.

- If $|w^{(t)}| > \tau$, then $|w^{(t+1)}| = |w^{(t)}| - \tau$.
- If $|w^{(t)}| \leq \tau$, then $w^{(t+1)} = 0$ (reaching optimum).

Thus, given any initial $w^{(0)}$, choosing $T = \lceil |w^{(0)}|/\tau \rceil$ yields

$$\forall t \geq T : \quad \left|w^{(t)}\right| = 0.$$



Hierarchy of Fixed-Point Operators (for ℓ_2-norm)

non-expansive

$$\|\mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}'\|_2 \leq \|\mathbf{w} - \mathbf{w}'\|_2 \quad \forall \mathbf{w}, \mathbf{w}' \in \mathbb{R}^d$$

Fixed-point iterations: stable, but may oscillate.

$\mathcal{F}\mathbf{w} = \mathcal{T}^{(f,\eta)}(\mathbf{w})$,
$\eta \in \left(\frac{1}{\beta}, \frac{2}{\beta}\right)$

$\mathcal{F} =$ rotation
$\mathcal{F} =$ reflection

firmly non-expansive

$$\|\mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}'\|_2^2 \leq \langle \mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}', \mathbf{w} - \mathbf{w}' \rangle$$

Converges to a fixed-point (if exists; generally **not unique***).*

$\mathcal{F}\mathbf{w} = \mathcal{T}^{(f,\eta)}(\mathbf{w})$,
$\eta \in \left(0, \frac{1}{\beta}\right]$

$\mathcal{F}\mathbf{w} = \mathbf{prox}_{\eta f(\cdot)}(\mathbf{w})$

$\mathcal{F} =$ projection

$\mathcal{F}\mathbf{w} = \mathbf{w}$

contraction

$$\|\mathcal{F}\mathbf{w} - \mathcal{F}\mathbf{w}'\|_2 \leq \kappa \|\mathbf{w} - \mathbf{w}'\|_2, \ \kappa < 1$$

$\mathcal{F}\mathbf{w} = \mathbf{c}$

Converges to **unique** *fixed-point.*

$\mathbf{prox}_{\eta f(\cdot)}(\cdot)$ and $\mathcal{T}^{(f,\eta)}$ (latter: $\eta \in (0, \frac{2}{\beta})$)
if f is σ-strongly convex

$\mathcal{F}\mathbf{w} = -0.5\mathbf{w}$

$\beta > 0$: β-smoothness of f; $\sigma > 0$: σ-strong convexity.

References

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- N. Parikh and S. Boyd, "Proximal Algorithms," *Foundations and Trends in Optimization*, vol. 1, no. 3, pp. 127–239, 2014.
- S. Bubeck, "Convex Optimization: Algorithms and Complexity," *Foundations and Trends in Machine Learning*, vol. 8, no. 3-4, pp. 231–357, 2015.

Further Resources

- **YouTube:** @alexjung111
- **LinkedIn:** Alexander Jung
- **GitHub:** alexjungaalto

Bonus Material

Strong Monotonicity of the Gradient

For σ-strongly convex functions f, it holds for all $\mathbf{w}, \mathbf{w}' \in \mathbb{R}^d$:

$$\|\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')\|_2 \geq \sigma \|\mathbf{w} - \mathbf{w}'\|_2 \tag{1}$$

- If f is σ-strongly convex, then by swapping $\mathbf{w}$ and $\mathbf{w}'$:

$$f(\mathbf{w}') \geq f(\mathbf{w}) + \langle \nabla f(\mathbf{w}), \mathbf{w}' - \mathbf{w} \rangle + \frac{\sigma}{2}\|\mathbf{w}' - \mathbf{w}\|_2^2$$

$$f(\mathbf{w}) \geq f(\mathbf{w}') + \langle \nabla f(\mathbf{w}'), \mathbf{w} - \mathbf{w}' \rangle + \frac{\sigma}{2}\|\mathbf{w} - \mathbf{w}'\|_2^2$$

- By adding both inequalities, rearranging and simplifying:

$$\langle \nabla f(\mathbf{w}) - \nabla f(\mathbf{w}'), \mathbf{w} - \mathbf{w}' \rangle \geq \sigma \|\mathbf{w} - \mathbf{w}'\|_2^2.$$

- Next, applying the Cauchy-Schwarz inequality yields:

$$\|\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')\|_2 \cancel{\|\mathbf{w} - \mathbf{w}'\|_2}$$

$$\geq \langle \nabla f(\mathbf{w}) - \nabla f(\mathbf{w}'), \mathbf{w} - \mathbf{w}' \rangle \geq \sigma \|\mathbf{w} - \mathbf{w}'\|_2^{\cancel{2}}$$

Details: Analysing the Gradient Step

Lemma (Co-Coercivity [4, Lemma 3.11])

*If f is convex and β-smooth, then ∇f is $1/\beta$-**co-coercive**:*

$$\langle \nabla f(\mathbf{w}) - \nabla f(\mathbf{w}'), \mathbf{w} - \mathbf{w}' \rangle \geq \frac{1}{\beta} \|\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')\|_2^2.$$

Analyse the operator $\mathcal{T}^{(f,\eta)} : \mathbb{R}^d \to \mathbb{R}^d$, $\mathbf{w} \mapsto \mathbf{w} - \eta \nabla f(\mathbf{w})$:

$$\begin{aligned}
&\left\|\mathcal{T}^{(f,\eta)}(\mathbf{w}) - \mathcal{T}^{(f,\eta)}(\mathbf{w}')\right\|_2^2 \\
=&\|(\mathbf{w} - \mathbf{w}') - \eta(\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}'))\|_2^2 \\
=&\|\mathbf{w} - \mathbf{w}'\|_2^2 + \eta^2 \|\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')\|_2^2 \\
&- 2\eta \langle \mathbf{w} - \mathbf{w}', \nabla f(\mathbf{w}) - \nabla f(\mathbf{w}') \rangle \\
\leq&\|\mathbf{w} - \mathbf{w}'\|_2^2 - \eta \left(\frac{2}{\beta} - \eta \right) \|\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')\|_2^2 \qquad (2)
\end{aligned}$$

Details: Analysing the Gradient Step (ctd.)

- If $\eta \in (0, 2/\beta]$, the term $\eta(2/\beta - \eta)$ is non-negative, i.e.

$$\left\|\mathcal{T}^{(f,\eta)}(\mathbf{w}) - \mathcal{T}^{(f,\eta)}(\mathbf{w}')\right\|_2^2 \leq \|\mathbf{w} - \mathbf{w}'\|_2^2,$$

  and in this case $\mathcal{T}^{(f,\eta)}$ is a non-expansive operator.
- If f is additionally σ-strongly convex, combine (2), (1) for

$$\begin{aligned}&\left\|\mathcal{T}^{(f,\eta)}(\mathbf{w}) - \mathcal{T}^{(f,\eta)}(\mathbf{w}')\right\|_2^2 \\ &\leq \left(1 - \eta\sigma^2\left(\frac{2}{\beta} - \eta\right)\right)\|\mathbf{w} - \mathbf{w}'\|_2^2,\end{aligned}$$

  and since $\kappa(\eta) := 1 - \eta\sigma^2(2/\beta - \eta) < 1$ for $\eta \in (0, 2/\beta)$, the operator $\mathcal{T}^{(f,\eta)}$ is a contractive operator in this case; note that $\kappa(\eta)$ becomes minimal for $\eta = 1/\beta$.

Details: Analysing the Gradient Step (ctd.)

Equivalent to $\mathcal{T}^{(f,\eta)}$ being a firmly non-expansive operator is

$$\langle \mathcal{T}^{(f,\eta)}(\mathbf{w}) - \mathcal{T}^{(f,\eta)}(\mathbf{w}'), (\mathbf{w} - \mathbf{w}') - (\mathcal{T}^{(f,\eta)}(\mathbf{w}) - \mathcal{T}^{(f,\eta)}(\mathbf{w}')) \rangle \geq 0.$$

- For f a β-smooth convex function, using co-coercivity,

$$\begin{aligned}
&\langle \mathcal{T}^{(f,\eta)}(\mathbf{w}) - \mathcal{T}^{(f,\eta)}(\mathbf{w}'), (\mathbf{w} - \mathbf{w}') - (\mathcal{T}^{(f,\eta)}(\mathbf{w}) - \mathcal{T}^{(f,\eta)}(\mathbf{w}')) \rangle \\
=&\langle (\mathbf{w} - \mathbf{w}') - \eta(\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')), \eta(\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')) \rangle \\
=&\eta \langle \mathbf{w} - \mathbf{w}', \nabla f(\mathbf{w}) - \nabla f(\mathbf{w}') \rangle - \eta^2 \|\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')\|_2^2 \\
\geq&\frac{\eta}{\beta} \|\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')\|_2^2 - \eta^2 \|\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')\|_2^2 \\
=&\eta \left(\frac{1}{\beta} - \eta \right) \|\nabla f(\mathbf{w}) - \nabla f(\mathbf{w}')\|_2^2
\end{aligned}$$

- For $\eta \in (0, 1/\beta]$, we find that $\eta(1/\beta - \eta)$ is ≥ 0; in this case $\mathcal{T}^{(f,\eta)}$ is even firmly non-expansive operator.